MSG No. 192.250 $P6/mc'c'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `62'2'`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,2,3,4,5,6,13,14,15,16,17,18]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[7,8,9,10,11,12,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `6/m..`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `4c`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[1,3,5,16,17,18]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[2,4,6,13,14,15]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[7,9,11,22,23,24]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[8,10,12,19,20,21]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$	`[1,3,5,8,10,12]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$	`[2,4,6,7,9,11]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$	`[16,17,18,19,20,21]`

Table 5: Wyckoff site: `4e`, site symmetry: `6..`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,2,3,4,5,6]`
2	$[0,\, 0,\, -z]$	`[7,8,9,10,11,12]`
3	$\left[0,\, 0,\, \frac{1}{2} - z\right]$	`[13,14,15,16,17,18]`
4	$\left[0,\, 0,\, z + \frac{1}{2}\right]$	`[19,20,21,22,23,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, \frac{1}{4}\right]$	`[1,4,13,17]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{4}\right]$	`[2,5,15,16]`
3	$\left[0,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[3,6,14,18]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[7,10,19,23]`
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[8,11,21,22]`
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,12,20,24]`

Table 7: Wyckoff site: `6g`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,7,10]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,5,8,11]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,6,9,12]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,17,19,23]`
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[14,18,20,24]`
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[15,16,21,22]`

Table 8: Wyckoff site: `8h`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,3,5]`
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[2,4,6]`
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7,9,11]`
4	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[8,10,12]`
5	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	`[13,14,15]`
6	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[16,17,18]`
7	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	`[19,20,21]`
8	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[22,23,24]`

Table 9: Wyckoff site: `12i`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[2,5]`
3	$\left[0, \frac{1}{2}, z\right]$	`[3,6]`
4	$\left[\frac{1}{2}, 0, -z\right]$	`[7,10]`
5	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[8,11]`
6	$\left[0, \frac{1}{2}, -z\right]$	`[9,12]`
7	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[13,17]`
8	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[14,18]`
9	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[15,16]`
10	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[19,23]`
11	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[20,24]`
12	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[21,22]`

Table 10: Wyckoff site: `12j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,13]`
2	$\left[x,\ x,\ \frac{1}{4}\right]$	`[2,16]`
3	$\left[0,\ x,\ \frac{1}{4}\right]$	`[3,14]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,17]`
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,18]`
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7,19]`
8	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[8,22]`
9	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[9,20]`
10	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[10,23]`
11	$\left[x,\ x,\ \frac{3}{4}\right]$	`[11,21]`
12	$\left[0,\ x,\ \frac{3}{4}\right]$	`[12,24]`

Table 11: Wyckoff site: `12k`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,17]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,15]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,18]`
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[4,13]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,16]`
6	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[6,14]`
7	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[7,23]`
8	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[8,21]`
9	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[9,24]`
10	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[10,19]`
11	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[11,22]`
12	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[12,20]`

Table 12: Wyckoff site: `12l`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,10]`
2	$[x-y,\ x,\ 0]$	`[2,11]`
3	$[-y,\ x-y,\ 0]$	`[3,12]`
4	$[-x,\ -y,\ 0]$	`[4,7]`
5	$[-x+y,\ -x,\ 0]$	`[5,8]`
6	$[y,\ -x+y,\ 0]$	`[6,9]`
7	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[13,23]`
8	$\left[y,\ x,\ \frac{1}{2}\right]$	`[14,24]`
9	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[15,22]`
10	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[16,21]`
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[17,19]`
12	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[18,20]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[x-y,\, x,\, z]$	`[2]`
3	$[-y,\, x-y,\, z]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[-x+y,\, -x,\, z]$	`[5]`
6	$[y,\, -x+y,\, z]$	`[6]`
7	$[-x,\, -y,\, -z]$	`[7]`
8	$[-x+y,\, -x,\, -z]$	`[8]`
9	$[y,\, -x+y,\, -z]$	`[9]`
10	$[x,\, y,\, -z]$	`[10]`
11	$[x-y,\, x,\, -z]$	`[11]`
12	$[-y,\, x-y,\, -z]$	`[12]`
13	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[y,\, x,\, \frac{1}{2}-z\right]$	`[14]`
15	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	`[15]`
16	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	`[16]`
17	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	`[17]`
18	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	`[18]`
19	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	`[19]`
20	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[20]`
21	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	`[21]`
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	`[22]`
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	`[23]`
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[24]`